Project4: Interim Data Analysis Plan

Eduardo Romo

May 07, 2026

Introduction

Linear regression is widely used to model the relationship between an outcome variable and a set of predictors. In moderate- to high-dimensional settings, variable selection becomes critical for accurately quantifying these relationships. Variable selection refers to identifying a subset of relevant predictors to include in a statistical model. This is important for both statisticians and collaborators, as including irrelevant variables can increase variance, lead to overfitting, and reduce interpretability, while excluding important variables can introduce bias and result in incorrect inference. A variety of variable selection methods have been developed, differing in both their procedures and performance. Classical approaches, such as backward selection, rely on stepwise regression techniques that begin with a full model and iteratively remove variables based on criteria such as p-values, Akaike Information Criterion (AIC), or Bayesian Information Criterion (BIC). In contrast, regularization methods such as lasso and elastic net incorporate penalties into the loss function to control model complexity. Lasso applies an ℓ_1 penalty that can shrink coefficients exactly to zero, thereby performing variable selection, while elastic net combines ℓ_1 and ℓ_2 penalties to better handle correlated predictors. Despite their widespread use, the performance of these methods can vary depending on factors such as sample size, correlation structure among predictors, and the strength of the signal. In this simulation study, we systematically compare several variable selection techniques in linear regression under controlled settings. Because the true underlying model is known, simulation allows direct evaluation of each method's ability to correctly identify relevant predictors, avoid selecting irrelevant variables, and accurately estimate regression coefficients. We focus on settings with 20 predictors, of which 5 have true associations with the outcome, and evaluate how performance varies across different sample sizes, correlation structures, and regularization choices.

Methods

We conducted a simulation study to evaluate the performance of several variable selection methods in linear regression. Data were generated from a Guassian linear model of the form $Y = X\beta + \epsilon$, where $X \in \mathbb{R}^{n\times 20}$ is a matrix of predictors and $\epsilon \sim N(0, \sigma^2)$. The true coefficient vector β contained five nonzero coefficients corresponding to the first five predictors, with values (0.5/3,1/3,1.5/3,2/3,2.5/3), while the remaining fifteen coefficients were set to zero. Predictors were generated using the hdrm::gen_data() function with an exchangeable correlation structure, where all pairwise correlations were equal to ρ. We considered three correlation settings ($\rho = 0, 0.35, 0.7$) to represent independent, moderately correlated, and highly correlated predictors. Two sample sizes (n = 250 and n = 500) were examined, resulting in six simulation scenarios. For each scenario, 10,000 independent datasets were generated. We compared five variable selection approaches: backward selection based on p-values, backward selection using AIC, backward selection using BIC, lasso regression ($\alpha = 1$), and elastic net regression ($\alpha = 0.5$). For the penalized regression methods, the tuning parameter λ was selected using 10-fold cross-validation, considering both the value that minimized the cross-validated error (λ_{min}) and the conservative one standard error rule (λ_{1se}). All models were fit without an intercept. After variable selection, the final model for each method was refit using ordinary least squares including only the selected predictors. From these refitted models, we extracted coefficient estimates, p-values, and

95% confidence intervals. Although this approach ignores uncertainty introduced by the selection process, it reflects common practice and allows us to assess the impact of model selection on subsequent inference. Performance was evaluated in terms of both selection and inference. Selection performance was assessed using the proportion of simulations in which each predictor was selected, allowing us to estimate true positive rates for the five relevant predictors and false positive rates for the fifteen noise predictors. Inference performance was evaluated among selected variables using bias of the estimated coefficients, coverage of 95% confidence intervals, and hypothesis testing metrics. Specifically, type I error was defined as the proportion of times noise variables were both selected and found to be statistically significant, while power was defined as the proportion of times true predictors were selected and statistically significant. All simulations were implemented in R using the hdrm, glmnet, and stats packages, with parallel computation used to improve efficiency.

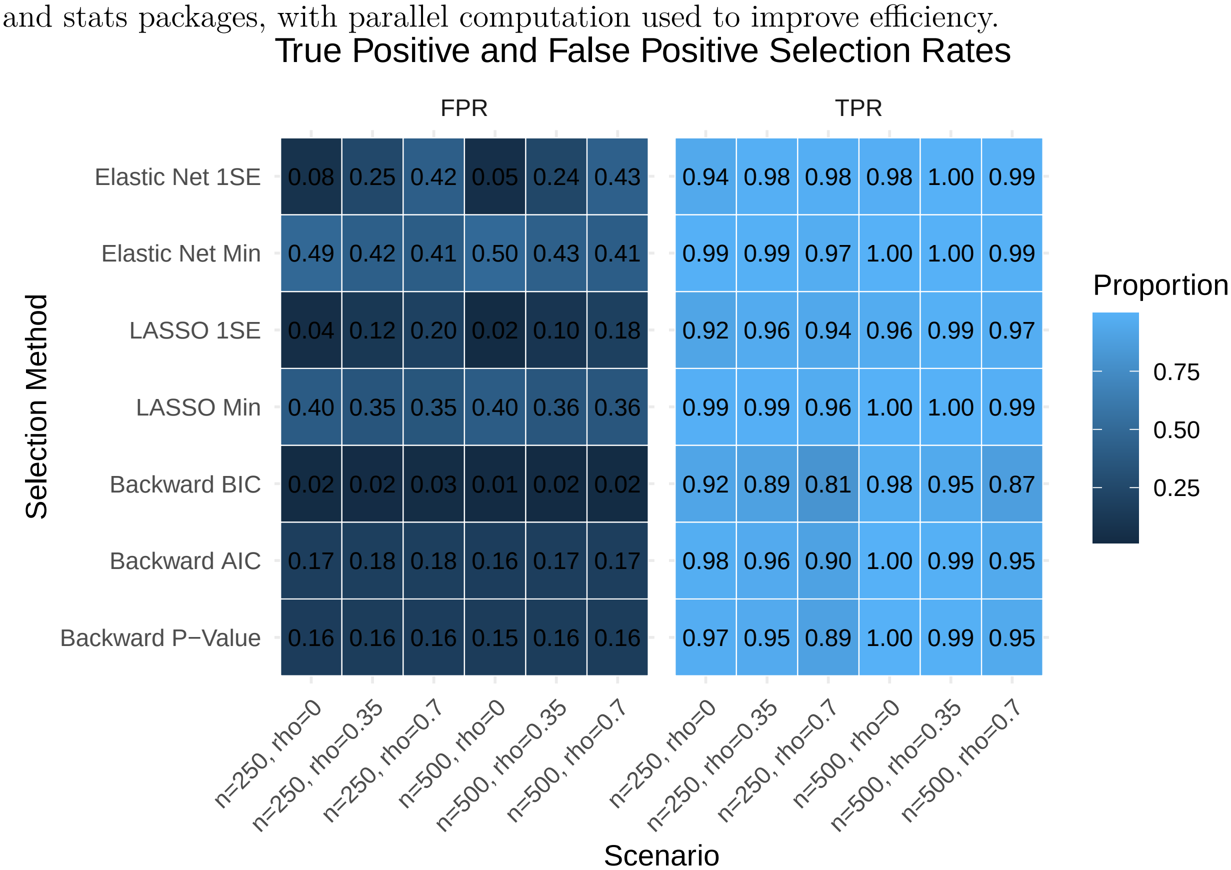

Table 1: Type II error (X1–X5) and Type I error (X6–X20) across scenarios.

		$n = 250$						$n = 500$					
ρ	Method	X_1	X_2	X_3	X_4	X_5	X_6–X_{20}	X_1	X_2	X_3	X_4	X_5	X_6–X_{20}
$\rho = 0$	Backward P-Value	0.257	0.001	0.000	0	0	0.055	0.042	0.000	0	0	0	0.052
	Backward AIC	0.257	0.001	0.000	0	0	0.055	0.042	0.000	0	0	0	0.052
	Backward BIC	0.388	0.003	0.000	0	0	0.021	0.115	0.000	0	0	0	0.013
	LASSO Min	0.270	0.001	0.000	0	0	0.052	0.046	0.000	0	0	0	0.051
	LASSO 1SE	0.422	0.008	0.000	0	0	0.028	0.180	0.000	0	0	0	0.017
	Elastic Net Min	0.270	0.001	0.000	0	0	0.052	0.046	0.000	0	0	0	0.051
	Elastic Net 1SE	0.326	0.002	0.000	0	0	0.040	0.090	0.000	0	0	0	0.034
$\rho = 0.35$	Backward P-Value	0.394	0.010	0.000	0	0	0.062	0.126	0.000	0	0	0	0.059
	Backward AIC	0.394	0.010	0.000	0	0	0.062	0.126	0.000	0	0	0	0.059
	Backward BIC	0.530	0.018	0.000	0	0	0.024	0.240	0.000	0	0	0	0.016
	LASSO Min	0.535	0.025	0.000	0	0	0.045	0.208	0.000	0	0	0	0.045
	LASSO 1SE	0.534	0.022	0.000	0	0	0.013	0.189	0.000	0	0	0	0.014
	Elastic Net Min	0.531	0.025	0.000	0	0	0.049	0.204	0.000	0	0	0	0.050
	Elastic Net 1SE	0.584	0.030	0.000	0	0	0.010	0.239	0.000	0	0	0	0.009
$\rho = 0.7$	Backward P-Value	0.649	0.144	0.007	0	0	0.065	0.411	0.012	0	0	0	0.061
	Backward AIC	0.652	0.146	0.007	0	0	0.065	0.412	0.013	0	0	0	0.061
	Backward BIC	0.763	0.200	0.011	0	0	0.027	0.607	0.026	0	0	0	0.018
	LASSO Min	0.809	0.292	0.022	0	0	0.034	0.597	0.036	0	0	0	0.035
	LASSO 1SE	0.831	0.301	0.023	0	0	0.008	0.617	0.036	0	0	0	0.009
	Elastic Net Min	0.806	0.292	0.023	0	0	0.038	0.590	0.035	0	0	0	0.040
	Elastic Net 1SE	0.852	0.331	0.027	0	0	0.006	0.650	0.043	0	0	0	0.007

Table 2: Bias and empirical 95% confidence interval coverage among selected true predictors.

			X_1		X_2		X_3		X_4		X_5	
ρ	n	Method	Bias	Coverage	Bias	Coverage	Bias	Coverage	Bias	Coverage	Bias	Coverage
$\rho = 0$	$n = 250$	Backward P-Value	0.016	0.971	0.001	0.943	0.000	0.945	0.000	0.944	0.000	0.945
		Backward AIC	0.015	0.970	0.001	0.943	0.000	0.945	0.000	0.944	0.000	0.946
		Backward BIC	0.040	0.959	0.002	0.949	0.000	0.950	0.000	0.945	0.000	0.949
		LASSO Min	0.007	0.972	-0.001	0.944	-0.002	0.944	-0.002	0.943	-0.002	0.944
		LASSO 1SE	0.038	0.958	0.000	0.952	-0.002	0.948	-0.002	0.946	-0.002	0.948
		Elastic Net Min	0.005	0.969	0.000	0.944	-0.002	0.944	-0.002	0.944	-0.002	0.945
		Elastic Net 1SE	0.026	0.967	-0.001	0.947	-0.002	0.948	-0.002	0.946	-0.003	0.948
	$n = 500$	Backward P-Value	0.001	0.958	0.000	0.947	0.000	0.950	0.000	0.948	-0.001	0.952
		Backward AIC	0.001	0.957	0.000	0.947	0.000	0.949	0.000	0.949	-0.001	0.952
		Backward BIC	0.010	0.970	0.000	0.949	0.000	0.952	0.000	0.951	-0.001	0.953
		LASSO Min	0.000	0.950	0.000	0.946	0.000	0.949	0.000	0.948	-0.002	0.951
		LASSO 1SE	0.013	0.969	0.000	0.948	0.000	0.952	0.000	0.950	-0.001	0.953
		Elastic Net Min	0.000	0.948	0.000	0.947	0.000	0.949	0.000	0.949	-0.002	0.952
		Elastic Net 1SE	0.006	0.971	0.000	0.949	0.000	0.952	0.000	0.950	-0.002	0.952
$\rho = 0.35$	$n = 250$	Backward P-Value	0.031	0.959	0.002	0.940	0.002	0.937	0.002	0.939	0.000	0.935
		Backward AIC	0.029	0.959	0.002	0.939	0.002	0.937	0.002	0.939	0.000	0.934
		Backward BIC	0.063	0.945	0.011	0.948	0.009	0.938	0.009	0.936	0.007	0.938
		LASSO Min	-0.014	0.976	-0.024	0.930	-0.024	0.930	-0.023	0.933	-0.025	0.925
		LASSO 1SE	-0.001	0.986	-0.024	0.933	-0.024	0.928	-0.024	0.933	-0.026	0.924
		Elastic Net Min	-0.015	0.969	-0.023	0.933	-0.022	0.932	-0.022	0.935	-0.024	0.928
		Elastic Net 1SE	-0.022	0.983	-0.034	0.928	-0.034	0.925	-0.034	0.930	-0.036	0.921
	$n = 500$	Backward P-Value	0.006	0.968	-0.001	0.941	0.001	0.940	0.000	0.939	0.001	0.941
		Backward AIC	0.005	0.968	-0.001	0.940	0.001	0.940	0.000	0.939	0.001	0.940
		Backward BIC	0.021	0.963	0.002	0.943	0.004	0.940	0.004	0.940	0.004	0.940
		LASSO Min	-0.016	0.940	-0.017	0.931	-0.016	0.933	-0.016	0.928	-0.016	0.933
		LASSO 1SE	-0.011	0.968	-0.017	0.932	-0.015	0.934	-0.016	0.930	-0.015	0.934
		Elastic Net Min	-0.015	0.940	-0.016	0.935	-0.015	0.936	-0.015	0.931	-0.015	0.935
		Elastic Net 1SE	-0.023	0.940	-0.025	0.924	-0.024	0.928	-0.024	0.925	-0.024	0.927
$\rho = 0.7$	$n = 250$	Backward P-Value	0.083	0.934	0.017	0.961	0.005	0.934	0.004	0.932	0.005	0.934
		Backward AIC	0.080	0.936	0.016	0.961	0.004	0.934	0.003	0.933	0.004	0.934
		Backward BIC	0.150	0.829	0.053	0.945	0.025	0.930	0.021	0.923	0.022	0.925
		LASSO Min	-0.013	0.984	-0.047	0.927	-0.048	0.924	-0.048	0.923	-0.047	0.926
		LASSO 1SE	-0.005	0.991	-0.053	0.933	-0.056	0.918	-0.057	0.916	-0.056	0.917
		Elastic Net Min	-0.016	0.983	-0.045	0.929	-0.045	0.927	-0.046	0.927	-0.045	0.928
		Elastic Net 1SE	-0.036	0.993	-0.059	0.927	-0.058	0.927	-0.059	0.926	-0.058	0.928
	$n = 500$	Backward P-Value	0.033	0.960	0.002	0.942	0.000	0.936	0.002	0.940	0.002	0.935
		Backward AIC	0.032	0.960	0.002	0.942	0.000	0.935	0.002	0.940	0.001	0.936
		Backward BIC	0.073	0.927	0.018	0.947	0.012	0.925	0.014	0.930	0.014	0.925
		LASSO Min	-0.023	0.974	-0.032	0.925	-0.034	0.921	-0.032	0.927	-0.033	0.923
		LASSO 1SE	-0.021	0.992	-0.038	0.918	-0.040	0.915	-0.039	0.921	-0.039	0.919
		Elastic Net Min	-0.023	0.967	-0.030	0.927	-0.032	0.926	-0.030	0.931	-0.031	0.928
		Elastic Net 1SE	-0.036	0.954	-0.041	0.924	-0.042	0.923	-0.040	0.929	-0.040	0.927